

## ATP-dependent thermoring basis for the heat unfolding of the first nucleotide-binding domain isolated from human CFTR.

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Traditionally, the thermostability of a protein is defined by a melting temperature, at which half of the protein is unfolded. However, this definition cannot indicate the structural origin of a heat-induced unfolding pathway. Here, the thermoring structures were studied on the ATP-dependent heat-induced unfolding of the first nucleotide-binding domain from the human cystic fibrosis transmembrane conductance regulator. The results showed that initial theoretical and experimental melting thresholds aligned well after three structural perturbations including the F508del mutation, the most common cause of cystic fibrosis. This alignment further demonstrated that the heat-induced unfolding process began with the disruption of the least-stable noncovalent interaction within the biggest thermoring along the single peptide chain. The C-terminal region, which was related to the least-stable noncovalent interaction and the ATP-dependent dimerization of two nucleotide-binding domains, emerged as a crucial determinant of the thermal stability of the isolated protein and a potential interfacial drug target to alleviate the thermal defect caused by the F508del mutation. This groundbreaking discovery significantly advances our understanding of protein activity, thermal stability, and molecular pathology.

## Trikafta rescues F508del-CFTR by tightening specific phosphorylation-dependent interdomain interactions.

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Trikafta effectively corrects the thermal and gating defects associated with the F508del mutation, the most common cause of cystic fibrosis, even at physiological temperatures. However, the exact correction pathway is still unclear. Here, noncovalent interactions among two transmembrane domains (TMD1 and TMD2), the regulatory (R) domain and two nucleotide binding domains (NBD1 and NBD2) were analyzed. The thermal stability of NBD1 was also evaluated through its tertiary constrained noncovalent interaction networks or thermoring structures. The results demonstrated that Trikafta binding to flexible TMD1 and TMD2 rearranged their interactions with the R domain upon phosphorylation, coupling tightened cytoplasmic TMD1-TMD2 interactions to tightened Mg/ATP-dependent NBD1-NBD2 dimerization, which stabilized NBD1 above human body temperature. In essence, while the F508 deletion primarily causes a thermal defect in NBD1, leading to a gating defect at the TMD1-TMD2 interface, Trikafta allosterically reverses these effects. These mechanistic insights into the precise correction pathway of this misfolded channel facilitate optimizing cystic fibrosis treatment. (155 words). Trikafta binding to flexible TMD1 and TMD2 tightened their cytoplasmic interactions. Tight cytoplasmic TMD1-TMD2 interactions primed the specific binding of the dynamic phosphorylated S813 site to the TMD1/TMD2/NBD1 interfaces. The tight binding of the S813 site to the TMD1/TMD2/NBD1 interfaces strengthened NBD1-NBD2 dimerization which stabilizes NBD1.

## Patient and family issues regarding genetic testing for cystic fibrosis: a review of prenatal carrier testing and newborn screening.

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Cystic fibrosis (CF) is a potentially life-shortening autosomal recessive genetic condition resulting in chronic progressive respiratory involvement, malnutrition, electrolyte imbalance, and male infertility. It is the most common autosomal inherited condition in the white population, and its presence is recorded with varying prevalence across ethnicities. Since the 1989 discovery of the genetic variant F508del, the most common cystic fibrosis transmembrane conductance regulator (CFTR) mutation, more than 1900 CF mutations have been identified. The 1997 National Institutes of Health (NIH) Consensus Statement on Cystic Fibrosis, along with 2001 and 2005 recommendations from the American College of Obstetricians and Gynecologists (ACOG), provide the basis for population CF carrier screening in the prenatal setting. Recommendations for newborn screening (NBS) for cystic fibrosis were released in 2004, with NBS programs in the United States initiated thereafter. With the wide variety of CFTR mutations and mutation combinations, there is not a clear understanding of the genotype-phenotype correlations or of the anticipated clinical trajectory for an individual who has identified CFTR mutations. This ambiguity creates challenges for patients and families in decision making related to CFTR carrier screening during the prenatal period, understanding the results of newborn screening for CF, or coping with the new genetic knowledge obtained. This literature review examines research regarding genetic testing for CF as it related to population screening. Patient and family issues from both the prenatal period and newborn testing are reviewed. Opportunities for future nursing research and implication for nursing practice are discussed.

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## Cystic Fibrosis Mutation Spectrum in North Macedonia: A Step Toward Personalized Therapy.

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The most prevalent "rare" disease worldwide, cystic fibrosis (CF), is an autosomal recessive multisystem disease, caused by mutations in the CFTR gene. The knowledge of CFTR mutations present in certain population is important for designing a simple, fast and cost-effective genetic testing approach, also for better management of CF patients, including the administration of novel targeted therapies. Here, we present genetic results of 158 unrelated CF patients from the National CF Registry of the Republic of North Macedonia. Initially, patients were screened for the 11 most common CF mutations. Additional CF mutations and large deletions/duplications in the CFTR gene were analyzed using commercial kits. If the genotype was undetermined, all CFTR exons were analyzed using Sanger DNA sequencing or next generation sequencing (NGS) (since 2014). The most common CF mutation, c.1521\_1523del (legacy name F508del), was found with an overall incidence of 75.9%. Additionally, 26 other pathogenic variants and three large deletions were identified in the CFTR gene as a genetic cause of CF. Two of these, c.1070 C>T (p.Ala357Val) and c.2779\_2788dup CTTGCTATGG (p.Gly930AlafsTer48), were novel. According to the distribution and prevalence of the pathogenic variants detected in our patients, a fast and cost-effective method, based on a single base extension was designed as a first-line CF genetic test with a 90.0% detection rate within our population. Furthermore, the knowledge of CFTR mutation classes in our CF patients represents the first step toward personalized therapy for CF in our country.

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## Phenotypic Evaluation of Rare Cystic Fibrosis Transmembrane Conductance Regulator Mutation Combinations in People with Cystic Fibrosis in Queensland, Australia.

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**Background:** Cystic fibrosis (CF) is a multisystem disorder caused by mutations in the cystic fibrosis transmembrane conductance regulator (CFTR) gene. We describe the distribution of CFTR mutation profiles in sub-tropical Queensland, Australia, and characterise the phenotypes associated with 'rare' CFTR mutation combinations. **Methods:** We conducted a retrospective observational study to analyse the CFTR mutation profiles of 322 people with CF (pwCF) under the care of a large adult CF centre in Queensland, Australia. Molecular pathology results were available for all identifiable CFTR mutations. The CFTR2 database was utilised to characterise the less common CFTR mutations to define mutation classes and explore associated phenotypic sequelae. **Results:** In total, eighty-seven different genotypes were identified within our CF cohort, with the most abundant mutation being the F508del mutation, 298/322 (92.5%). Thirty-six pwCF with CFTR mutations are considered to have 'rare' CFTR mutations, and eleven with previously undefined phenotypes. For these eleven pwCF, late diagnosis in adulthood was confirmed in 5/11 pwCF (45.5%) with CFTR modulator therapy only initiated in 5/11 (45.5%). **Conclusions:** The profile of more common CFTR genotypes within our cohort of adult pwCF living in Queensland, Australia, generally reflects the global predominance of F508del, G542X, G551D, N1303K, and R117H. The phenotypic heterogeneity of disease seen within the eleven pwCF in our cohort with previously undefined CFTR genotypes highlights that rare mutations can also be associated with severe disease and continue to be at risk of delayed diagnosis. Access to CFTR modulator therapies for this group of pwCF remains limited and should remain a research priority.

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## The common variant of cystic fibrosis transmembrane conductance regulator is recognized by hsp70 and degraded in a pre-Golgi nonlysosomal compartment.

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The most common cause of cystic fibrosis is deletion of Phe-508 (delta F508) from the cystic fibrosis transmembrane conductance regulator (CFTR). Previous studies have suggested that delta F508 CFTR is an unstable protein that retains a pattern of glycosylation specific to the endoplasmic reticulum. This report examines the mechanism responsible for the mislocalization of delta F508 CFTR in a human cystic fibrosis epithelial cell line overexpressing recombinant CFTR by virtue of adenovirus-mediated gene transfer. Immunoelectron microscopy confirmed that wild-type CFTR is delivered to the plasma membrane of these cells and that delta F508 CFTR is retained in the endoplasmic reticulum. Pulse-chase studies showed that newly synthesized CFTR complexes with the chaperone hsp70. The wild-type protein dissociates from hsp70 before its transport to the Golgi, and the protein is subsequently degraded in lysosomes. By contrast, the complex formed between delta F508 CFTR and hsp70 is retained in the endoplasmic reticulum and delta F508 CFTR is rapidly degraded in a pre-Golgi nonlysosomal compartment. Thus, hsp70 discriminates between the normal form of CFTR and the form of the protein that most commonly causes cystic fibrosis (delta F508). These findings clarify the mechanism by which mutation causing delta F508 affects the intracellular trafficking of CFTR and suggest another function for hsp70 in ensuring quality control during the biosynthesis of plasma-membrane proteins.

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## Genotype and phenotype in cystic fibrosis.

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Cystic fibrosis (CF) is caused by mutations in the CF transmembrane conductance regulator (CFTR) gene

which encodes a protein expressed in the apical membrane of exocrine epithelial cells. CFTR functions principally as a cAMP-induced chloride channel and appears capable of regulating other ion channels. Besides the most common mutation, DeltaF508, accounting for about 70% of CF chromosomes worldwide, more than 850 mutant alleles have been reported to the CF Genetic Analysis Consortium. These mutations affect CFTR through a variety of molecular mechanisms which can produce little or no functional CFTR at the apical membrane. This genotypic variation provides a rationale for phenotypic effects of the specific mutations. The extent to which various CFTR alleles contribute to clinical variation in CF is evaluated by genotype-phenotype studies. These demonstrated that the degree of correlation between CFTR genotype and CF phenotype varies between its clinical components and is highest for the pancreatic status and lowest for pulmonary disease. The poor correlation between CFTR genotype and severity of lung disease strongly suggests an influence of environmental and secondary genetic factors (CF modifiers). Several candidate genes related to innate and adaptive immune response have been implicated as pulmonary CF modifiers. In addition, the presence of a genetic CF modifier for meconium ileus has been demonstrated on human chromosome 19q13.2. The phenotypic spectrum associated with mutations in the CFTR gene extends beyond the classically defined CF. Besides patients with atypical CF, there are large numbers of so-called monosymptomatic diseases such as various forms of obstructive azoospermia, idiopathic pancreatitis or disseminated bronchiectasis associated with CFTR mutations uncharacteristic for CF. The composition, frequency and type of CFTR mutations/variants parallel the spectrum of CFTR-associated phenotypes, from classic CF to mild monosymptomatic presentations. Expansion of the spectrum of disease associated with the CFTR mutant genes creates a need for revision of the diagnostic criteria for CF and a dilemma for setting nosologic boundaries between CF and other diseases with CFTR etiology.

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## Cystic fibrosis mutations: report from the French Registry. The Clinical Centers of the CF.

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Data from 2,666 patients with cystic fibrosis (CF) born in France, submitted during the period of 1992-1996 to the French registry for CF, were used to describe the different mutations, their frequency and their regional distribution. A total of 5,332 CF chromosomes have been analyzed, demonstrating 229 different mutations and accounting for 87% of CF genes in the French population. DeltaF508 is the most common mutation at 67.9% of CF mutations, followed by G542X (2.5%), N1303K (2.0%), 1717-1G-->A (1.2%), R553X (0.8%) and G551D (0.7%). The data show a clear geographical variation in the distribution of many of the mutations. Given the geographical heterogeneity of these mutations, carrier screening does not appear to be feasible in most French regions.

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## Application of multiplex ARMS and SSCP/HD analysis in molecular diagnosis of cystic fibrosis in Indian patients.

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Cystic fibrosis (CF) is an autosomal recessive disorder caused by mutations in the CFTR gene. The most severe, DeltaF508, mutation accounts for nearly 70% of CF cases worldwide. Besides DeltaF508, there are other point mutations, namely G542X, G551D, R553X, N1303K, and 621+1(G-->T), which are

common among Caucasians. Additionally, a polyT polymorphism in intron 8 is also involved in the pathogenesis of CF. However, neither the prevalence nor the types of mutations causing CF in India are known. In this study, we aimed at estimating the frequency of the above common mutations and polymorphism in clinically suspected CF cases. We also carried out partial analysis of the CFTR gene, limited to exons 10 and 11, to identify other variations in these exons. The multiplex amplification refractory mutation system (ARMS) test was applied for rapid simultaneous analysis of six most common CF mutations, in 100 normal and 39 elevated sweat chloride cases. The scanning of exons 10 and 11 was carried out by single-stranded conformation polymorphism/heteroduplex (SSCP/HD) analysis, followed by DNA sequencing in 50 normal and 37 elevated sweat chloride cases. A single ARMS-polymerase chain reaction assay was used to distinguish the 5T, 7T, and 9T alleles in 100 normal and 33 elevated sweat chloride cases. The multiplex ARMS analysis identified the DeltaF508 mutation at an allele frequency of 24% in Indian CF cases. However, the other predominant CF mutations were found to be absent. The 7T polyT variant was observed to be the most common allele, followed by the 9T, and 5T, which was the lowest. The DeltaF508 mutation was observed in all instances with the 9T variant. The SSCP/HD and DNA sequencing additionally revealed a known polymorphism (M470V, exon 10) and a known mutation [1525-1(G-->A), intron 9]. The 1525-1(G-->A) mutation, observed in a single 4-year-old male, is predicted to code for a class II defective CFTR protein. The findings of this study suggest a difference in relative frequencies and spectrum of CFTR mutations in Indian CF cases. A larger screening study of the entire CFTR gene in clinically well defined CF cases is required to delineate common mutations in the CFTR gene and enable molecular diagnosis of CF in India.

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## Targeting ubiquitination machinery in cystic fibrosis: Where do we stand?

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Cystic Fibrosis (CF) is a genetic disease caused by mutations in CFTR gene expressing the anion selective channel CFTR located at the plasma membrane of different epithelial cells. The most commonly investigated variant causing CF is F508del. This mutation leads to structural defects in the CFTR protein, which are recognized by the endoplasmic reticulum (ER) quality control system. As a result, the protein is retained in the ER and degraded via the ubiquitin-proteasome pathway. Although blocking ubiquitination to stabilize the CFTR protein has long been considered a potential pharmacological approach in CF, progress in this area has been relatively slow. Currently, no compounds targeting this pathway have entered clinical trials for CF. On the other hand, the emergence of Orkambi initially, and notably the subsequent introduction of Trikafta/Kaftrio, have demonstrated the effectiveness of molecular chaperone-based therapies for patients carrying the F508del variant and even showed efficacy against other variants. These treatments directly target the CFTR variant protein without interfering with cell signaling pathways. This review discusses the limits and potential future of targeting protein ubiquitination in CF.